Examiner only

(iii) The equation for a straight line is:

$y = mx + c$

where:

$m$ = gradient
$y$ = velocity of blood flow
$x$ = lumen diameter
$c$ = $y$-intercept of the graph

On this graph, $c = 0$. Calculate the velocity of blood flow in a vessel with a lumen diameter of 160 μm using your calculated value of $m$ and the equation above. [1]

Velocity of blood flow = .................................................. au

(iv) In the single circulatory system of a fish, oxygenated blood leaves the capillaries of the gill lamellae and passes to the systemic circulation. Explain the importance of the relationship shown by the graph in the return of deoxygenated blood through veins to the fish's heart. [2]

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